VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama" Belagavi, Karnataka- 590018

INTERNSHIP REPORT

ON

"CELL SENESCENCE DETECTION & DNA ISOLATION TECNIQUES FROM PLANT AND ANIMAL MODELS"

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF

BACHELOR OF ENGINEERING

In

BIOTECHNOLOGY

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CERTIFICATE

This is to certify The Internship Report entitled **"CELL SENESCENCE DETECTION & DNA ISOLATION TECNIQUES FROM PLANT AND ANIMAL MODELS"** is a Bonafide record of the work carried out by Ms. ANUSHA (4GM21BT003) of 8th semester under the guidance and supervision at BASIC SCIENCE RESEARCH CENTRE , KLE BELGAVI in the partial fullfilment and of the requirements for the award of degree of Bachelor of Engineering in Biotechnology of Visvesvaraya Technological University belgavi during the academic year 2024-2025.

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Professor	Professor	Professor and Head	GM Institute of Technology

Name of Examiners	Signature of Examiners
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2	2

DECLARATION

We hereby declare that this Internhip Report entitled "**CELL SENESCENCE DETECTION & DNA ISOLATION TECNIQUES FROM PLANT AND ANIMAL MODELS**" is an Internship conducted under the guidance of Dr. Vijay Kumbar, Scientist Grade-I, BSRC,KLE Belgavi, submitted in partial fulfillment of the requirements of the degree of B.E BIOTECHNOLOGY of Visvesvaraya Technological University, Belagavi during the year 2024-2025.

I further declare that this work has not formed the basis for the award of any degree / diploma of any other university/institution.

Ms. Anusha (4GM21BT003)

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ABSTRACT

Cellular senescence is a critical biological process characterized by stable cell cycle arrest, which plays a dual role in tumor suppression and tissue aging. In the internship we investigated cellular senescence using a commercially available senescence detection kit based on senescence-associated β-galactosidase (SA-β-gal) activity. This allowed for effective identification and quantification of senescent cells in mammalian cultures. Further, genomic DNA was isolated using the TRIzol reagent, a lab based method used for isolation of DNA, Additionally, we also isolated DNA from plant tissues using a silica membrane-based spin column kit. This kit-based method provided high-quality DNA with minimal contamination, suitable for PCR and other molecular applications. Together, these protocols support comprehensive studies of senescence at both cellular and molecular levels in diverse biological systems, contributing to the understanding of aging and related pathologies.

Keywords: *Cellular senescence, SA-β-gal assay, TRIzol reagent, DNA isolation*, *Spin column kit*

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CHAPTER 1 ABOUT THE INSTITUTE

Dr. Prabhakar Kore
Basic Science Research Center
A Research Facility at KLE Academy of Higher Education and Research.
(Deemed to be University)

Fig 1. BSRC, Belgavi

The Dr. Prabhakar Kore Basic Science Research Center was established in the year 2012 with the aim of providing a platform for research to KAHER constituent units.

The center has 10000 sq ft laboratory space that houses Analytical Laboratory, Microbiological Laboratory, Cell culture laboratory and Molecular Biology and bioinformatics laboratory.

These laboratories are engaged in key areas like phytochemicals, drug delivery, molecular biology and genetics, microbiology, cell culture and cancer biology. Besides investigator driven research by the faculty the centre provides support for research by students and faculty of the constituent units of KAHER. Besides research, the center also conducts a Master of Science in Biotechnology course under KAHER.

The center also provides opportunities to young and enthusiastic students with training opportunities through internship programs. The center strives to contribute to the health of individuals through basic and applied research.

CHAPTER 2 : LIST OF PROJECTS TO BE CONDUCTED IN 3-MONTHS:

1.1 MOLECULAR BIOLOGY DEPARTMENT

The Molecular Biology Department at BSRC is conducting two major projects focused on DNA isolation from plant and animal cells.

Isolation of DNA from Plant Cells (Kit Based Method)

This project involves the extraction and purification of DNA from various plant species. The study aims to optimize the isolation process, ensuring high yield and purity, which are crucial for downstream applications such as PCR, sequencing, and genetic engineering. By studying different plant tissues, researchers will analyze the impact of factors like cell wall composition, secondary metabolites, and extraction techniques on DNA quality.

Isolation of DNA from Animal Cells (Laboratory Method)

This project focuses on developing efficient methodologies for extracting genomic DNA from animal tissues and cells. The extracted DNA will be used for genetic studies, disease diagnostics, and biotechnological applications. The research will explore different sample sources, including blood, muscle, and epithelial cells, to determine the most effective techniques for high-quality DNA isolation.

Both of these projects play a crucial role in advancing molecular biology techniques and can be applied to genetic research, biotechnology, and medical diagnostics.

1.2 CELL CULTURE DEPARTMENT

Cellular senescence SA-gal Marker based analysis . It is a critical biological process characterized by stable cell cycle arrest, which plays a dual role in tumor suppression and tissue aging. During our internship, in the **Cell Culture Department**, we studied cellular senescence using a commercially available kit-based method that detects senescence-associated β-galactosidase (SA-β-gal) activity. This assay enabled effective identification and quantification of senescent cells in mammalian cell cultures.

1.3 REVIEW ARTICLE ON “ EXOMES”

CHAPTER 3 : ISOLATION OF DNA FROM PLANT CELLS

PLANT SELECTED:

Ficus religiosa. L **(Peepal Tree)**

Fig 2: *Ficus religiosa*

Taxonomic Classification:

Kingdom: Plantae **Clade:** Tracheophytes **Clade:** Angiosperms **Clade:** Eudicots **Clade:** Rosids **Order:** Rosales **Family:** Moraceae **Genus:** *Ficus*

Subgenus: *F. subg. Urostigma*

Species: *Ficus religiosa*

Binomial Name: *Ficus religiosa* L.

Ficus religiosa is a large deciduous or semi-evergreen tree, growing up to 30 meters tall with a trunk diameter of 3 meters. It has cordate leaves with an elongated drip tip, measuring 10–17 cm in length. The tree produces syconium-type fruits, 1–1.5 cm in diameter, which ripen from green to purple. The tree contributes to biodiversity by providing habitat and sustenance for various insects, birds, and microorganisms, playing a vital role in ecosystems.

FLOW CHART METHODOLOGY:

20 mg Plant Sample

↓

Add 400μL API Buffer + 4 μL RNase Sonicate & Incubate for 10 min at 94°C

↓

Add 130μL Buffer P3, Mix and Incubate on ice

↓

Centrifuge for 5 min at 12000rpm

↓

Transfer the Supernatant to QIA shredder spin column

↓

Centrifuge 14000 rpm for 2 min

↓

Transfer the flow through to fresh micro-centrifuge tube

↓

Add 800 μL of Buffer AW1 & Mix by pipetting

↓

Transfer the mixture to DNA Minispin column & Centrifuge at 8000rpm for 1 min

↓

Discard the flow through but keep the spin column

↓

Add 500 μL Buffer AW2 to spin column & centrifuge at 8000rpm for 1 min & Discard the flow through

↓

Add another 500 μL Buffer of AW2, centrifuge at 14000 rpm for 2 min and discard the column tube with the flow through

↓

Transfer the spin column to a clean 1.5ml micro-centrifuge tube

↓

Add another 100 μL Buffer of AE into the center of Membrane and Incubate for 5min at room temperature

Centrifuge 8000 rpm for 1 min Flow through contains DNA

↓

Add another 100 μL Buffer of AE to the same column and incubate for 5 MIN centrifuge at 8000rpm for 1min

↓

Observe for the peak of 480nm under Nanophotometer

PROCEDURE:

Step 1: Sample Preparation and Lysis

Fig 3: Powdered *Ficus religiosa* Leaves sonicated with lysis buffer

1. Take **20 mg of plant sample** and place it in a microcentrifuge tube.
2. Add **400 µL of API Buffer** to the tube (Lysis Buffer)
3. Add **4 µL of RNase** to the sample to degrade RNA contamination.
4. Perform **sonication** to break the cell walls and release cellular contents.

Step 2: Precipitation and Centrifugation

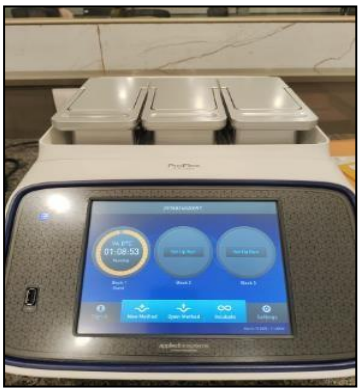


Fig 4: Incubation & centrifugation

1. **Incubate** the sample at **64°C for 10 minutes** to facilitate cell lysis.
2. Add **130 µL of Buffer P3** to the lysate.
3. Mix the solution thoroughly to ensure uniform dispersion of reagents.
4. **Incubate on ice** to enhance precipitation of unwanted cellular components.
5. **Centrifuge at 12,000 rpm for 5 minutes** to separate cellular debris from the supernatant.

Step 3: Purification Using QIA Shredder and Spin Column

1. Transfer the **supernatant** to a **QIA shredder spin column** for filtration.
2. **Centrifuge at 14,000 rpm for 2 minutes** to remove any residual debris.
3. Collect the flow-through in a fresh microcentrifuge tube.
4. Add **1.5 volumes of Buffer AW1** to the flow-through and mix thoroughly by pipetting.
5. Transfer the mixture to a **DNA Minispin column** and **centrifuge at 8,000 rpm for 1 min**
6. **Discard the flow-through** but **retain the spin column** for further purification.

Fig 5: QIA Shredder

Fig 6: Mini Spin column

Step 4: Washing, Elution, and DNA Quantification

1. Add **500 µL of Buffer AW2** to the spin column.
2. **Centrifuge at 8,000 rpm for 1 minute** and discard the flow-through.
3. Add another **500 µL of Buffer AW2**, **centrifuge at 14,000 rpm for 2 minutes**, and discard the column tube with the flow-through.
4. Transfer the spin column to a clean **1.5 mL microcentrifuge tube**.
5. Add **100 µL of AE Buffer** to the center of the membrane and **incubate at room temperature for 5 minutes**.
6. **Centrifuge at 8,000 rpm for 1 minute** to elute the DNA into the flow-through.
7. Repeat elution with another **100 µL of AE Buffer**, **incubate for 5 minutes**, and centrifuge at **8,000 rpm for 1 minute**.
8. Measure DNA concentration by observing the **peak at 480 nm using a Nanophotometer**
9. **UV spectroscopy** can be used for **DNA quantification**, primarily through **absorbance measurements at 260 nm**. This method is based on the principle that **nucleic acids absorb UV light** due to the presence of purine and pyrimidine bases.

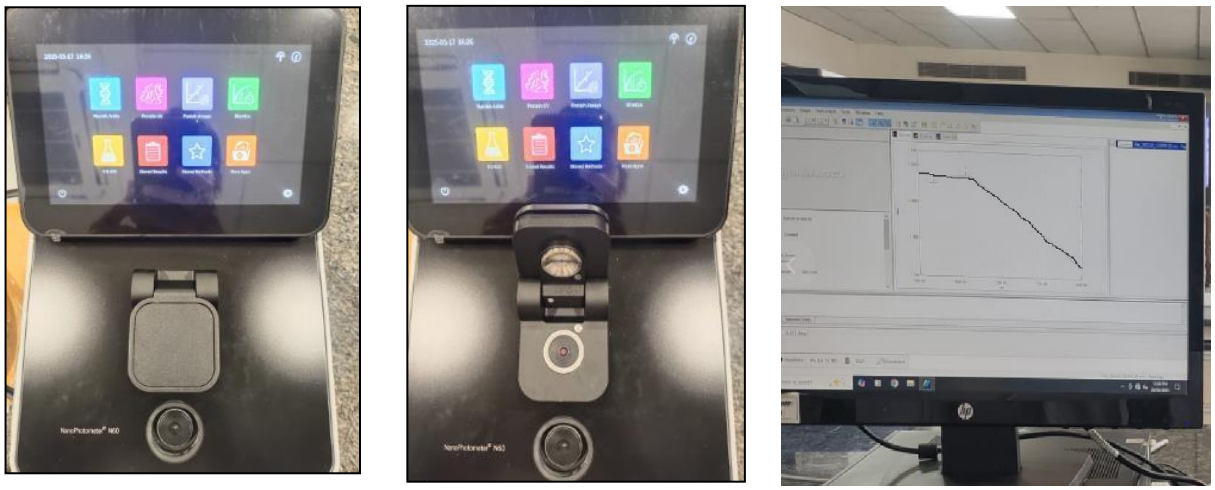


Fig 7: Nanophotometer & UV spectroscopy for DNA Quantification

Successfully demonstrated DNA extraction from plant cells to students Rani Channamma University (Belgavi)

CHAPTER 4: ISOLATION OF DNA FROM ANIMAL CELLS

To estimate the amount of DNA present in a sample using Trizol reagent.

FLOW CHART METHODOLOGY:

Collect Cell Culture

↓

Centrifuge at 2000 rpm

↓

Discard Supernatant (Keep Pellet)

↓

Add 200µL Trizol to the Pellet

↓

Homogenize (Sonication/Vortex)

↓

Incubate at Room Temperature (5-10 min)

↓

Add 40µL Chloroform & Shake 30 sec

↓

Incubate at Room Temperature (2-3 min)

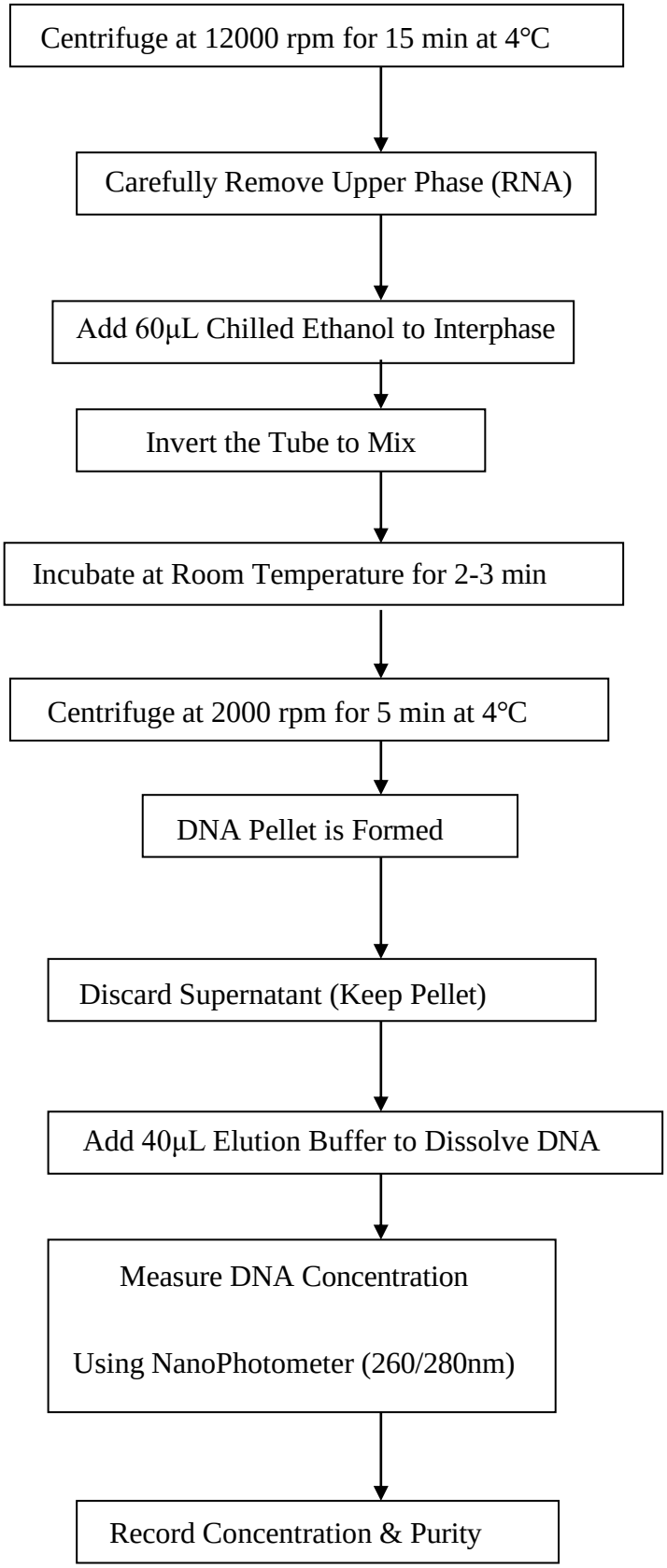
↓

PHASE SEPARATION

Lower Phase (Phenol-Chloroform) → Proteins

Middle Phase (Interphase) → DNA

Upper Phase (Aqueous) → RNA


Centrifuge at 12000 rpm for 15 min at 4°C
Carefully Remove Upper Phase (RNA)
Add 60µL Chilled Ethanol to Interphase
Invert the Tube to Mix
Incubate at Room Temperature for 2-3 min
Centrifuge at 2000 rpm for 5 min at 4°C
DNA Pellet is Formed
Discard Supernatant (Keep Pellet)
Add 40µL Elution Buffer to Dissolve DNA
Measure DNA Concentration
Using NanoPhotometer (260/280nm)
Record Concentration & Purity

PROCEDURE:

Step 1: Sample Preparation

1. **Take the suspended cell culture** in a centrifuge tube.
2. **Centrifuge it at 2000 rpm for 5 minutes** to pellet the cells.
3. **Carefully discard the supernatant** (the liquid part) without disturbing the cell pellet.
4. **Note:** Ensure that the pellet is not lost while discarding the supernatant.

Fig 8 : Cells centrifuged at 2000rpm

Step 2: Add Trizol & Homogenization

1. **Add 200µL of Trizol reagent** to the cell pellet.
2. **Homogenize the mixture** either by:
   - **Sonication** (for 30 seconds to 1 minute) OR
   - **Vortexing** the tube until the pellet dissolves completely.
3. **Incubate the tube at room temperature for 5-10 minutes** to allow complete dissociation of nucleoproteins.

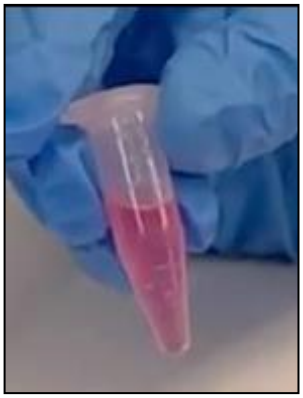


Fig: Trizol incubated with the cells

Note: Avoid overheating the sample during sonication.

Step 3: Chloroform Addition & Phase Separation

1. **Add 40µL of chloroform** to the sample containing Trizol.
2. **Close the tube tightly** and shake it vigorously for **15-30 seconds** to mix the contents well.
3. **Incubate the mixture at room temperature for 2-3 minutes.**
4. **Centrifuge the sample at 12,000 rpm for 10-15 minutes at 4°C.**

Note: Always use a **refrigerated centrifuge** for proper phase separation.

Fig 9: Addition of Chloroform and Centrifugation

After centrifugation, you will observe three layers:

- **Lower red phenol-chloroform layer** – Contains proteins.
- **Middle interphase** – Contains DNA.
- **Upper aqueous phase** – Contains RNA.

Fig 10: Phase separation after centrifugation

Step 4: DNA Precipitation

1. **Carefully remove the upper aqueous phase** using a pipette without disturbing
2. To the **interphase (middle layer)**, **add 60µL of 100% chilled ethanol.**
3. **Mix the tube gently by inversion** (DO NOT vortex).
4. **Incubate at room temperature for 2-3 minutes.**
5. **Centrifuge the sample at 2000 rpm for 5 minutes at 4°C. Note:**

- **Always use chilled ethanol** to ensure proper precipitation of DNA.
- **Immediately centrifuge** after adding ethanol, otherwise the DNA might get denatured.

Fig11: White DNA at bottom of micro-centrifuge

6. **After centrifugation, a white pellet of DNA will be visible at the bottom.**
7. **Carefully discard the supernatant** without disturbing the DNA pellet.
8. **Add 40µL of elution buffer** to the pellet to dissolve the DNA.
9. **Gently pipette up and down** to ensure the DNA is completely dissolved.

Step 6: DNA Concentration Estimation Using Nanophotometer

1. **Take 1-2µL of elution buffer (Blank) in a nanophotometer** and calibrate it.
2. Then, take **1-2µL of your DNA sample** (dissolved in elution buffer) and measure the absorbance.
3. Peak seen at **260 nm and 280 nm** shows the presence of DNA.

Note:

- The **260/280 ratio** should ideally be around **1.8** for pure DNA.
- If the ratio is lower, it indicates protein contamination.
- If the ratio is higher, it indicates RNA contamination.

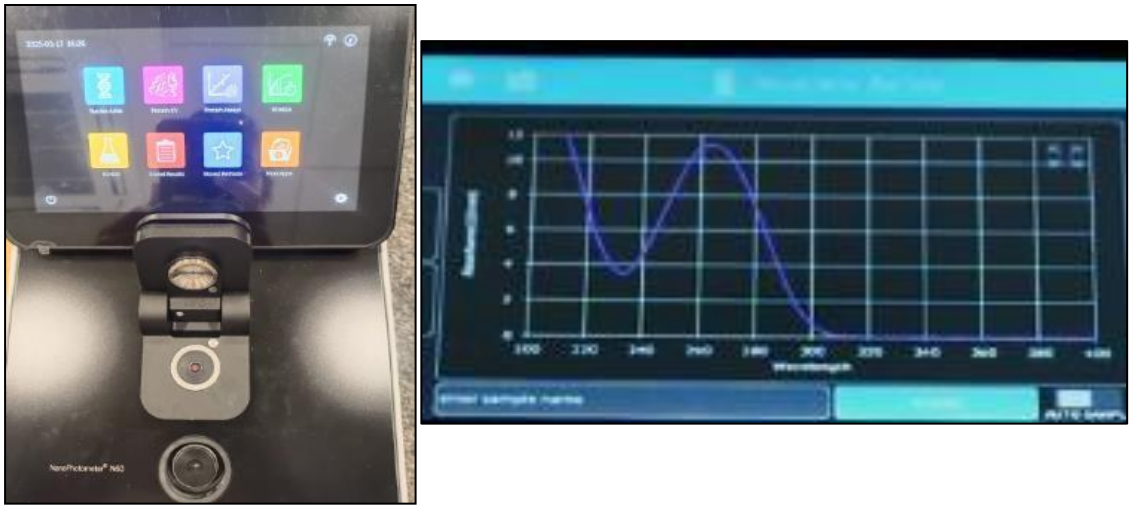


Fig 12 : Nanophotometer peak seen at about 260nm

Precaution Summary:

- Always use **chilled ethanol** for DNA precipitation.
- After adding ethanol, immediately centrifuge the sample to prevent DNA denaturation.
- Avoid disturbing the interphase layer during phase separation.
- Always use a refrigerated centrifuge for accurate separation.

CHAPTER 6: CELL SENESENCE DETECTION

AIM

To detect senescent cells in vitro using the EZdetect™ Cell Senescence Detection Kit based on senescence-associated β-galactosidase (SA-β-gal) activity at pH 6.0.

Fig 13: EZdetect™ Cell Senescence Detection Kit

PRINCIPLE

Cells are fixed to the culture plate so that they don't change in structure using fixing solution, then they are washed and stain is added , the stain contains X-gal which is a synthetic molecule that resembles the natural substrate for ß-galactosidase which is the lysosomal enzyme produced in the cell when it undergoes lysis. When ß-galactosidase is present , it hydrolyses X-gal , releaseing a blue coloured product, this blue coloured product forms a blue precipitate.

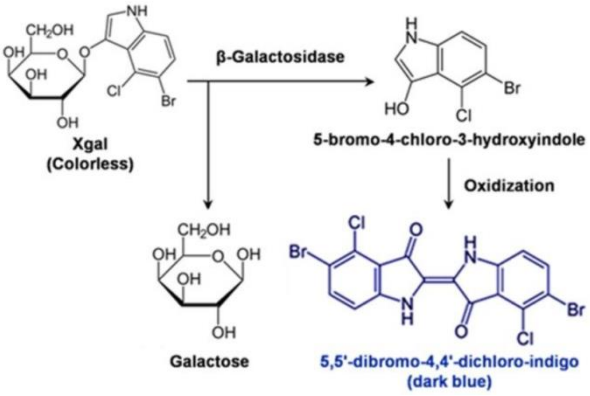


Fig 14: Activity of X-gal as substrate on ß-galactosidase

MATERIALS REQUIRED

PROVIDED IN THE KIT

- Staining solution A (CCK063A)
- Staining solution B (CCK063B)
- Staining solution C (CCK063C)
- X-gal solution (CCK063D)
- Fixing solution 10X (CCK063E)
- 10X Dulbecco's Phosphate Buffered Saline (DPBS) (CCK063F)

NOT PROVIDED IN THE KIT

- Adjustable pipettes and repeat pipettor
- Incubator (at 37°C, without CO_2)
- Inverted microscope
- Parafilm
- Glycerol
- Sterile cell culture grade water
- Polypropylene tubes
- Cell culture dishes (35 mm or others)

PROCEDURE

A. PREPARATION OF REAGENTS

1. **1X Fixing Solution**

- Dilute 10X fixing solution with sterile cell culture grade water in 1:9 ratio.
- Store at -20°C after use.

2. **1X DPBS**

- Dilute 10X DPBS (CCK063F) with sterile water in 1:9 ratio.
- Store at room temperature.

3. **X-gal Solution**

- Thaw at room temperature and warm at 37°C in a water bath for 1 hour before use.

Note: Prevent aggregates that can interfere with cell staining.

4. **STAINING MIXTURE PREPARATION**

Component	For 10 ml	For 2ml
Staining A	1 ml	200 µl
Staining B	125 µl	25 µl
Staining C	125 µl	25 µl
X-gal	250 µl	50 µl
Sterile Water	8.5 ml	1700 µl

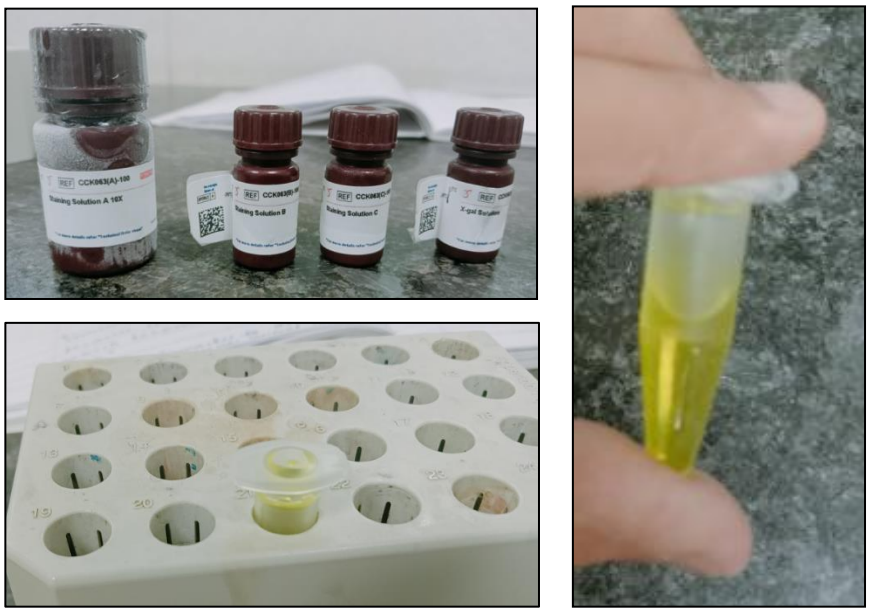


Fig 14: Prepared staining solution

B. STAINING PROTOCOL

1. Aspirate growth medium from cells.
2. Wash cells with 1 ml of 1X DPBS. Swirl gently and aspirate.
3. Repeat DPBS wash once more.
4. Add 1 ml of 1X Fixing solution. Incubate at room temp for 8–10 min.

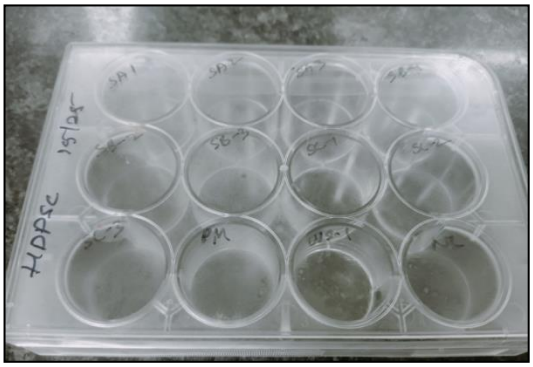
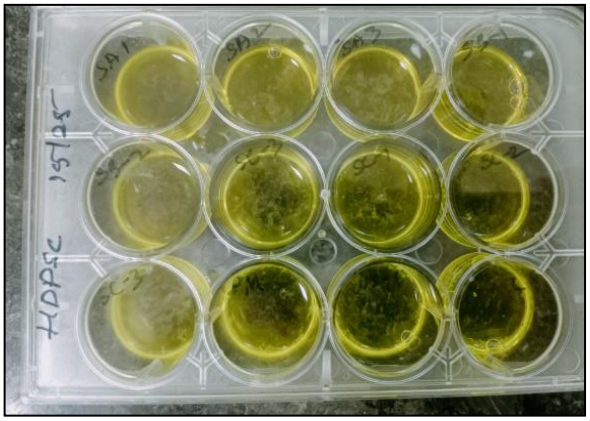


Fig 15: Cells washed with DPBS/PBS and fixed using fixing solution

During this, prepare the staining mixture as mentioned above

5. Aspirate fixing solution.
6. Wash twice with 1 ml of 1X DPBS.
7. Add 1 ml of staining mixture to the dish.
8. Incubate at 37°C *without* CO_2 for 2 hours or overnight.
   - **Note:** Do NOT use CO_2 incubator; pH sensitivity may affect results.



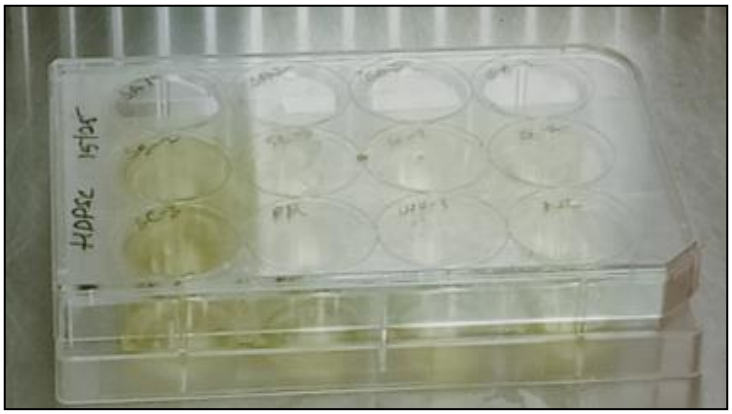


Fig 16: addition of the staining solution & Incubation

9. Observe under an inverted microscope.
10. Count blue-stained (senescent) and total cells.
11. Optional: Store stained plates with 70% glycerol overlay at 2–8°C.

OBSERVATION

- **Positive result**: Blue coloration (from cleaved X-gal) indicates SA-β-gal activity, signifying senescence.
- **Negative result**: No or faint blue staining.

Fig 17: Blue stain observed after leaving the cells overnight

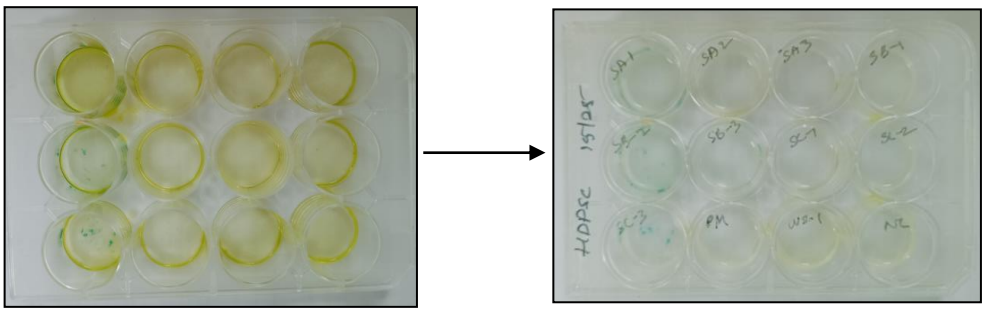


Fig 18: Aspiration of the staining solution and washing with PBS buffer saline

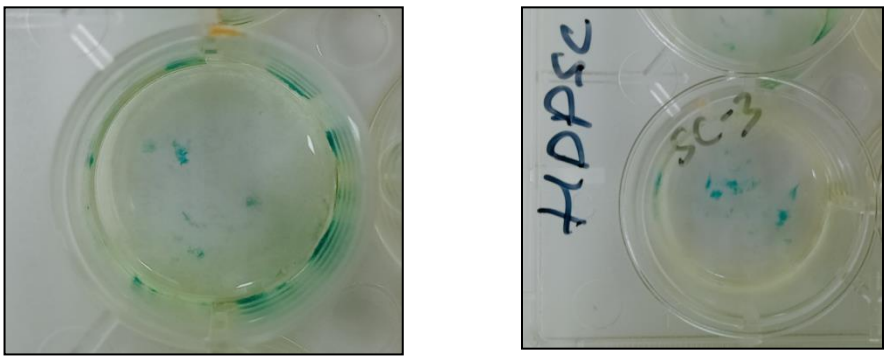


Fig 19: Blue stained cells showing positive results and SA- β-gal activity

RESULT

The assay confirms the presence of senescent cells in a given culture, indicated by SA-β-gal activity.

4) REVIEW ARTICLE ON EXOME BASED IMMUNOTHERAPY AGAINST CANCER

ABSTRACT

Cancer is the second leading cause of death globally. Although various cancer therapies have been developed, they are not sufficient and cannot completely cure cancer. Recently, there has been a trend towards utilizing exosomes in therapy development. Exosomes are extracellular vesicles (EVs) that contain miRNA, lipids, proteins, and lncRNA and others enclosed in a lipid bilayer. These molecules play a significant role in cancer progression, aiding the transition from benign to malignant tumors, and they can serve as critical biomarkers for cancer detection. Immune cells also produce exosomes, which can enhance immunotherapy. Various immune cells, such as dentritic cells (DCs), neutrophills,NK cells, macrophages, basophills, B cells, and CD4+ and CD8+ T cells, release exosomes that can be modified to increase their antigenicity. Among these immune responses, the CD8+ T cell response is the most robust. However, tumor cells have developed the ability to induce the expression of PDL-1 on T cells, rendering them ineffective and leading to T cell exhaustion. This has promoted research into anti-PDL-1 therapies, which are currently under investigation. In this review, we discuss the structure and components of exosomes, their potential as biomarkers, the effectiveness of immune cell exosomes, the mechanisms of CD8+ T cell exhaustion, anti-PDL-1 therapies, and modifications of exomes to act as drug delivery systems. We also explore exosome inhibitors that are being studied for their activity against cancer exosomes.

Keywords: ***exosomes , dentritic cells, neutrophills,NK cells, macrophages, basophills***

CONCLUSION:

Plant and animal DNA extraction, along with cell senescence detection, are powerful tools in modern biology. They help in improving agriculture, diagnosing diseases, protecting wildlife, studying aging, and developing better treatments for cancer. These techniques play an important role in research, healthcare, and environmental conservation, showing how science can solve real-world problems and improve lives.

REFERENCES:

1. G. Raposo, & W. Stoorvogel, Extracellular vesicles: Exosomes, microvesicles, and friends. *Journal of Cell Biology*,(2013) *200*(4), 373-383.
2. S. Anticoli, et al. Exosome-based strategies for infectious disease vaccines. *International Journal of Molecular Sciences*, (2018)*19*(7), 974.
3. Y. Cheng,& J. S. Schorey, Exosomes carrying mycobacterial antigens can induce protective immune responses against *Mycobacterium tuberculosis* infection. *Nature Communications*, (2013). *4*, 1710.
4. C. Thery, et al. Exosomes in immune responses: From immune surveillance to vaccine development. *Trends in Immunology*, (2009). *30*(12), 505-511.